



Exam Ace Champs

Where Champions Are Made

Quick Revision

Chapterwise

Mind-Maps

MATHEMATICS

class
11

Covers

all chapters of NCERT



Faster Recall



Quick Revision

Let A and B be two sets. If every element of A is an element of B, then A is called a subset of B and written as $A \subset B$ or $B \supset A$ (read as 'A' is contained in 'B' or 'B contains A'). B is called superset of A.

Note:

1. Every set is a subset and superset of itself.
2. If A is not a subset of B, we write $A \not\subset B$.
3. The empty set is the subset of every set.
4. If A is a set with $n(A) = m$, then no. of element A are 2^n and the number of proper subsets of A are $2^n - 1$.

Eg. Let $A = \{3, 4\}$, then subsets of A are ϕ , $\{3\}$, $\{4\}$, $\{3, 4\}$. Here, $n(A) = 2$ and number of subsets of A = $2^2 = 4$.

A set is collection of well-defined distinguished objects. The sets are usually denoted by capital letters A, B, C etc., and the members or elements of the set are denoted by lower case letters a, b, c etc.,. If x is a member of the set A, we write $x \in A$ (read as 'x' belongs to A) and if x is not a member of set A, we write $x \notin A$ (read as 'x' doesn't belong to A). If x and y both belong to A, we write $x, y \in A$. Some examples of sets are: A: odd numbers less than 10
N: the set of all rational numbers
B: the vowels in the English alphabates
Q: the set of all rational numbers.

Subset

Introduction

The number of elements in a finite set is represented by $n(A)$, known as cardinal number.
Eg.: $A = \{a, b, c, d, e\}$ Then, $n(A) = 5$

Cardinal Number

Sets

Types of Sets

Empty set or Null set

A set which has no element is called null set. It is denoted by symbol ϕ or $\{\}$.
E.g: Set of all real numbers whose square is -1.
In set-builder form: $\{x: x \text{ is a real number whose square is } -1\}$
In roaster form: $\{\}$ or ϕ

Equivalent set

Singleton set

Representation of Sets

Set builder form or Rule Method

Roaster or Tabular form

In this form, we write a variable (say x) representing any member of the set followed by property satisfied by each member of the set.
eg.: The set A of all prime number less than 10 in set builder form is written as
 $A = \{x \mid x \text{ is a prime number less than } 10\}$
The symbol " $\mid$ " stands for the word "such that". Sometimes, we use symbol ":" in place of symbol " $\mid$ "

In this form, we first list all the members of the set within braces (curly brackets) and separate these by commas. Eg: The set of all natural number less than 10 in this form is written as: $A = \{1, 3, 5, 7, 9\}$ In roaster form, every element of the set is listed only once. The order in which the elements are listed is immaterial. Eg. Each of the following sets denotes the same set $\{1, 2, 3\}$, $\{3, 2, 1\}$, $\{1, 3, 2\}$

Equal set

Two finite sets A and B are said to be equivalent, if $n(A) = n(B)$. Clearly, equal set are equivalent but equivalent set need not to be equal.
e.g.: The sets $A = \{4, 5, 3, 2\}$ and $B = \{1, 6, 8, 9\}$ are equivalent, but are not equal.

A set having one element is called singleton set.
e.g.: (i) $\{0\}$ is a singleton set, whose only member is 0.
(ii) $A = \{x: 1 < x < 3, x \text{ is a natural number}\}$ is a singleton set which has only one member which is 2.

Two sets A and B are set to be equal, written as $A = B$, if every element of A is in B and every element of B is in A.
e.g.: (i) $A = \{1, 2, 3, 4\}$ and $B = \{3, 1, 4, 2\}$, then $A = B$
(ii) $A = \{x: x-5=0\}$ and $B = \{x: x \text{ is an integral positive root of the equation } x^2 - 2x - 15 = 0\}$
Then $A = B$

Finite and Infinite set

A set which has finite number of elements is called a finite set.
Otherwise, it is called an infinite set.
E.g.: The set of all days in a week is a finite set whereas the set of all integers, denoted by $\{\dots, -2, -1, 0, 1, 2, \dots\}$ or $\{x \mid x \text{ is an integer}\}$ is an infinite set.
An empty set ϕ which has no element is a finite set A is called empty or void or null set.

Complement law:

(i) $A \cup A' = U$ (ii) $A \cap A' = \emptyset$ De Morgan's Law: (i) $(A \cup B)' = A' \cap B'$ (ii) $(A \cap B)' = A' \cup B'$ Double Complement law: $(A')' = A$ Law of empty set and universal set $\emptyset' = U$ and $U' = \emptyset$

The set of all subset of a given set A is called power set of A and denoted by $P(A)$. E.g : If $A = \{1,2,3\}$, then $P(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$. Clearly, if A has n elements, then its power set $P(A)$ contains exactly 2^n elements.

The set containing all objects of element and of which all other sets are subsets is known as universal sets and denoted by U .

E.g : For the set of all integers, the universal set can be the set of rational numbers or the set R of real numbers

If U is a universal set and A is a subset of U , then complement of A is the set which contains those elements of U , which are not present in A and is denoted by A' or A^c . Thus, $A^c = \{x: x \in U \text{ and } x \notin A\}$ e.g.: If $U = \{1, 2, 3, 4, \dots\}$ and $A = \{2, 4, 6, 8, \dots\}$ then $A^c = \{1, 3, 5, 7, \dots\}$

Properties of complement

Complement of set

Power set

Universal set

Sets

Operations on sets

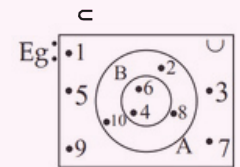
Algebra of sets

Venn Diagram

A Venn diagram is an illustration of the relationships between and among sets, groups of objects that share something common.

These diagrams consist of rectangle and closed curves usually circles

In the given venn diagram $U = \{1,2,3,\dots,10\}$ universe set of which $A = \{2,4,6,8,10\}$ and $B = \{4,6\}$ are subsets and also $B \subset A$



Symmetric Difference

Subsets of a set of real

Intersection

Disjoint sets

Union

The symmetric difference of two sets

A and B , denoted by $A \Delta B$, is defined as $(A \Delta B) = (A - B) \cup (B - A)$

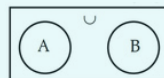
Eg. If $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 3, 5, 7, 9\}$ then $(A \Delta B) = (A - B) \cup (B - A) = \{2, 4\} \cup \{7, 9\} = \{2, 4, 7, 9\}$

The set of natural numbers $N = \{1, 2, 3, 4, 5, \dots\}$ The set of integers $Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

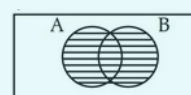
The set of irrational numbers, $T = \{x: x \in R \text{ and } x \notin Q\}$ The set of rational number $Q = \{x: x = \frac{p}{q}, p, q \in Z \text{ and } q \neq 0\}$ Relation among these subsets are $N \subset Z \subset Q$, $Q \subset R, T \subset R, N \not\subset T$

$$Q \subset R, T \subset R, N \not\subset T$$

Two sets A and B are said to be disjoint, if $A \cap B = \emptyset$ i.e. A and B have no common element. e.g: if $A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$ Then, $A \cap B = \emptyset$, so A and B are disjoint.



The intersection of two sets A and B , written as $A \cap B$ (read as 'A intersection B') is the set consisting of all the common elements of A and B . Thus, $A \cap B = \{x: x \in A \text{ and } x \in B\}$ Clearly, $x \in A \cap B \Rightarrow \{x \in A \text{ and } x \in B\}$ and $x \notin A \cap B \Rightarrow \{x \notin A \text{ or } x \notin B\}$. Eg: If $A = \{a, b, c, d\}$ and $B = \{c, d, e, f\}$ Then $A \cap B = \{c, d\}$



The union of two sets A and B , written as $A \cup B$ (read as 'A union B') is the set of all elements which are either in A or in B in both.

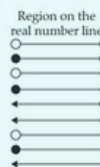
Thus, $A \cup B = \{x: x \in A \text{ or } x \in B\}$ clearly, $x \in A \cup B \Rightarrow x \in A \text{ or } x \in B$ and $x \notin A \cup B \Rightarrow x \notin A \text{ and } x \notin B$ eg: If $A = \{a, b, c, d\}$ and $B = \{c, d, e, f\}$ then $A \cup B = \{a, b, c, d, e, f\}$

Interval Notation

Let a and b be real numbers with $a < b$

Set of Real Numbers
• $\{x | a < x < b\}$
• $\{x | a \leq x < b\}$
• $\{x | a < x \leq b\}$
• $\{x | a \leq x \leq b\}$
• $\{x | x < a\}$
• $\{x | x \leq a\}$
• $\{x | x > a\}$
• $\{x | x \geq a\}$

Interval Notation
 (a, b)
 $[a, b)$
 $(a, b]$
 $[a, b]$
 $(-\infty, b)$
 $(-\infty, b]$
 $[a, \infty)$
 $[a, \infty]$
 $(-\infty, \infty)$



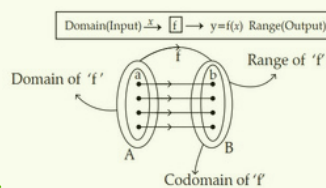
Let A & B be two empty sets. Then any subset ' R ' of $A \times B$ is a relation from A to B . If $(a, b) \in R$, then we write $a R b$, which is read as ' a is related to b ' by a relation R , ' b ' is also called image of ' a ' under R . The total number of relations that can be defined from a set A to a set B is the number of possible subsets of $A \times B$. If $n(A)=p$ and $n(B)=q$, then $n(A \times B)=pq$ and total number relations is 2^{pq} .

If R is a relation from A to B , then the set of first elements in R is called domain & the set of second elements in R is called range of R . Symbolically, Domain of $R = \{x: (x, y) \in R\}$; Range of $R = \{y: (x, y) \in R\}$. The set B is called co-domain of relation R . Note: the range $\subseteq$ Codomain.
Eg. Given, $R = \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6)\}$.
then Domain of $R = \{1, 2, 3, 4, 5\}$
Range of $R = \{2, 3, 4, 5, 6\}$ and codomain of $R = \{1, 2, 3, 4, 5, 6\}$

Given, $R = \{(x, y): x \text{ is the first letter of the name } y, x \in P, y \in Q\}$
Then, $R = \{(a, Ali), (b, Beena), (c, Chaitu)\}$ This is a visual or pictorial representation of relation R (called an arrow diagram) is shown in figure.

Given two non empty sets A & B . The cartesian product $A \times B$ is the set of all ordered pairs of elements from A & B i.e., $A \times B = \{(a, b) : a \in A; b \in B\}$.
If $n(A)=p$ and $n(B)=q$, then $n(A \times B)=pq$

Definition: A relation ' f ' from a set A to set B is said to be a function if every element of set A has one and only one image in set B .
Notations:



Let $f: X \rightarrow R$ and $g: X \rightarrow R$ be any two real functions where $X \subset R$.

Addition: $(f+g)(x) = f(x) + g(x); \forall x \in R$
Subtraction: $(f-g)(x) = f(x) - g(x); \forall x \in R$
Product: $(fg)(x) = f(x) \cdot g(x); \forall x \in R$
Quotient: $(f/g)(x) = f(x)/g(x);$ provided $g(x) \neq 0, \forall x \in R$

$f(x) = \log x, a > 0, a \neq 1$ Domain $= x \in (0, \infty)$
Range $= y \in R$

The function $f: R \rightarrow R$ defined by $y = f(x) = x \forall x \in R$ is called identity function. Domain $= R$ and Range $= R$

The function $f: R \rightarrow R$ defined by $y = f(x) = c, \forall x \in R$, where c is a constant is called constant function. Domain $= R$ and Range $= \{c\}$

Relation

Domain & range of a Relation

Pictorial representation of a relation

Inverse relation

Even and odd function

Relations and Functions

Functions

Algebra of function

Kinds of Functions

Related terms

Some standard real

Signum function

Greatest integer function

Exponential function

Log function

Identity function

Constant function

Modulus function

The function $f: R \rightarrow R$ defined by $f(x) = |x|$. Domain $= R$ and Range $= [0, \infty)$

$$\begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Even function
 $f(-x) = f(x), \forall x \in \text{Domain}$
Odd function
 $f(-x) = -f(x), \forall x \in \text{Domain}$

The function $f: R \rightarrow R$ defined by $f(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$ is called signum function. It is usually denoted by $y = f(x) = \text{sgn}(x)$ Domain $= R$ and Range $= \{0, -1, 1\}$

The function $f: R \rightarrow R$ defined by as the greatest integer less than or equal to x . It is usually denoted by $y = f(x) = [x]$. Domain $= R$ and Range $= Z$ (All integers)

$f(x) = ax, a > 0, a \neq 1$, Domain: $x \in R$ Range: $f(x) \in (0, \infty)$

Specific Instances to Generalisation

e.g.: Rohit eats food. Vikas eats food. Rohit and Vikas are men. Then all men eat food. Statement is true for $n=1, n=k$ and $n=k+1$, then, the statement is true for all natural numbers n .

Generalisation of Specific Instance

e.g.: Rohit is a man and all men eat food, therefore, Rohit eats food.

Induction

Deduction

Principle of
Mathematical
Induction

Examples

Steps for
Principle of
Mathematical
Induction Proof

Ex: Prove that $2n > n$ for all positive integer n .

Solution: **Step1:** Let $P(1): 2n > n$ **Step2:** When $n=1, 2 \times 1 > 1$. Hence $P(1)$ is true **Step3:** Assume that $P(k)$ is true for any positive integer $k, 2k > k$ **Step4:** We shall now prove that $P(k+1)$ is true. Multiplying both sides of step(3) by 2, we get $2 \cdot 2k > 2 \cdot k \Rightarrow 2k+1 > 2k \Rightarrow 2k+1 = k+k \Rightarrow 2k+1 > k+1$ (since $k > 1$) Therefore, $P(k+1)$ is true when $P(k)$ is true. Hence, by P.M.I., $P(n)$ is true for every positive integer n .

Step1: Let $P(n)$ be a result or statement formulated in terms of n in a given equation
Step2: Prove that $P(1)$ is true.
Step3: Assume that $P(k)$ is true.
Step4: Using step 3, prove that $P(k+1)$ is true.
Step5: Thus, $P(1)$ is true and $P(k+1)$ is true whenever $P(k)$ is true
Hence, by the principle of mathematical induction, $P(n)$ is true for all natural numbers n .

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Mathematical
Induction

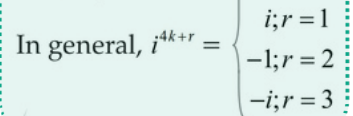
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$$-\pi < \theta \leq \pi.$$

$$\text{In general, } i^{4k+r} = \begin{cases} 1; r=1 \\ -1; r=2 \\ -i; r=3 \end{cases}$$

Conjugate of a complex number: For a given complex number $z=a+ib$, its conjugate is defined as $\bar{z}=a-ib$

- The notation $a < b$ means that 'a is less than b'.
- The notation $a > b$ means that 'a is greater than b'.
- The notation $a \geq b$ means that 'a is greater than or equal to b'.
- The notation $a \leq b$ means that 'a is less than or equal to b'.
- The notation $a \neq b$ means that 'a is not equal to b'.

Solution and Solution set

Solution: Values of x , which make inequality a true statement.

e.g.: '3' is the solution for $x < 7$.

Solution Set:

The set of values of x is called its solution set.

e.g.: $\{1, 2, 3, 4, 5, 6\}$ is solution set for $x < 7$, where x is a natural number.

Numerical Inequalities: Relation between numbers
e.g.: $3 < 5$, $4 > 7$
Literal or Variable Inequalities: Relation between two variables or variable and numbers.
e.g.: $y < 8$
Double Inequalities: Relationship from two sides
e.g.: $2 < x < 5$
Strict Inequalities: An inequality that uses the symbol $<$ or $>$. e.g.: $x < 5$, $3 < 5$
Slack Inequality: An inequality that uses the symbols $\leq$ or $\geq$. e.g.: $y \leq 5$
Linear Inequalities in One Variable: An inequality which involves a linear function in one variable
e.g.: $x < 5$
Linear Inequalities in two variables: An inequality which involves a linear function in two variables,
e.g.: $3x + 2y < 5$
Quadratic Inequalities: An inequality which involves a quadratic function, e.g.: $x^2 + 2x \leq 5$.

Solution of Linear Inequality in one Variable

Types of Inequalities

- An inequality is an relation that holds between two values, when they are different e.g.: $x < 5$, here, there is a relation between x & 5.
- Two real numbers or algebraic expressions related by symbol ' $<$ ', ' $>$ ', ' $\leq$ ' or ' $\geq$ ' form inequality.
e.g.: $y \leq 4$, $2x + 3y \geq 5$, $3 > x$.

Conditions

Graphical Solution of Linear Inequalities two variables

Linear Inequalities

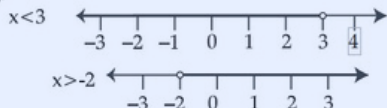
Notations

Representation

Rules of inequality

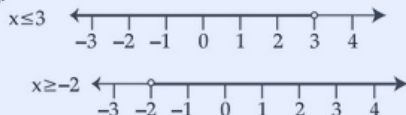
To represent $x < a$ ($x > a$) on a number line, put a circle on the number 'a' and dark line to the left (or right) of the number 'a'.

e.g:



To represent $x \leq a$ (or $x \geq a$) on a number line, put a dark circle on the number 'a' and dark the line to the left (or right) of the number 'a'.

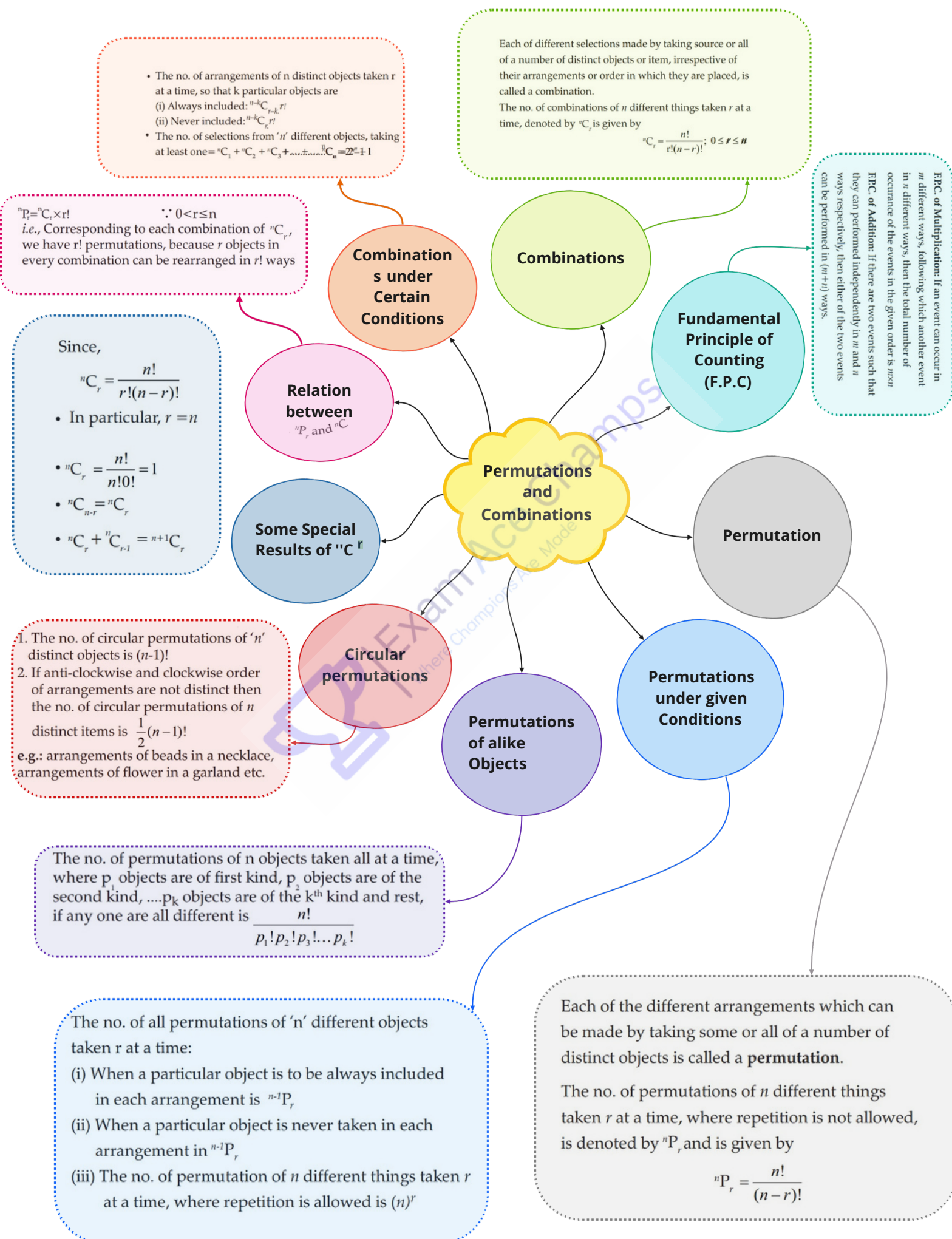
e.g:



- Equal numbers may be added to (or subtracted from) both the sides of an inequality without affecting the sign of inequality:
e.g.: $x < 7$ is same as $x + 2 < 7 + 2$
- Both sides of an inequality can be multiplied (or divided) by the same positive number without affecting the sign of inequality.
e.g.: $x + y < 7$ is same as $(x + y) \times 3 < 7 \times 3$
- When both sides are multiplied or divided by a -ve number, then sign of inequality is reversed.
e.g.: $x + y < 7$ is same as $(x + y) \times (-3) > 7 \times (-3)$

- A line divides the cartesian plane into two equal parts. A point in cartesian plane will either lie on a line or will lie in either of half planes. The region containing all the solutions of inequality is called the solution region.

- In order to identify the half plane represented by an inequality, it is just sufficient to take any point (a, b) (not on line) and check whether it satisfies the inequality or not. If it satisfies, then inequality represents the half plane and shade the region, which contain the point otherwise the inequality represents the half plane which does not contain the point within it.



If n is negative integer, then $n!$ is not defined. We state binomial theorem in another form

$$(a+b)^n = a^n + \frac{n}{1!}a^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 + \frac{n(n-1)(n-2)}{3!}a^{n-3}b^3 + \dots + \frac{n(n-1)(n-2)\dots(n-r+1)}{r!}a^{n-r}b^r + \dots + b^n$$

Here, $T_{r+1} = \frac{n(n-1)(n-2)\dots(n-r+1)}{r!}a^{n-r}b^r$

The general term of an expansion $(a+b)^n$ is

$$T_{r+1} = {}^nC_r a^{n-r} b^r, \quad 0 \leq r \leq n, r \in \mathbb{N}$$

Middle Terms:

1. In $(a+b)^n$, if n is even, then the no. of terms in the expansion is odd. Therefore, there is only one middle term and it is $\left(\frac{n+2}{2}\right)^{\text{th}}$ term.

2. In $(a+b)^n$, if n is odd then the no. of terms in the expansion is even.

Therefore, there are two middle terms and those are

$$\left(\frac{n+1}{2}\right)^{\text{th}} \text{ and } \left(\frac{n+3}{2}\right)^{\text{th}} \text{ terms.}$$

Binomial Theorem for any index

Some Special Cases $(a+b)^n$

General & Middle Terms in Expansion of $(a+b)^n$

Binomial Theorem

Binomial Coefficients

Definition

In the expansion of $(a+b)^n$,

(i) Taking $a=x$ and $b=-y$, we obtain

$$(x-y)^n = {}^nC_0 x^n - {}^nC_1 x^{n-1}y + {}^nC_2 x^{n-2}y^2 - \dots + (-1)^n {}^nC_n y^n$$

(ii) Taking $a=1$, $b=x$, we obtain

$$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + \dots + {}^nC_n x^n$$

(iii) Taking $a=1$, $b=-x$, we obtain

$$(1-x)^n = {}^nC_0 - {}^nC_1 x + {}^nC_2 x^2 - \dots + (-1)^n {}^nC_n x^n$$

The coefficient ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ in the expansion of $(a+b)^n$ are called binomial coefficients and denoted by $C_0, C_1, C_2, \dots, C_n$ respectively

Properties of binomial coefficients:

- $C_0 + C_1 + C_2 + \dots + C_n = 2^n$
- $C_0 - C_1 + C_2 - \dots + (-1)^n C_n = 0$
- $C_0 + C_2 + C_4 + \dots = C_1 + C_3 + C_5 + \dots = 2^{n-1}$
- ${}^nC_{r_1} = {}^nC_{r_2} \Rightarrow r_1 = r_2 \text{ or } r_1 + r_2 = n$
- ${}^nC_{r_1} + {}^nC_{r_1-1} = {}^{n+1}C_{r_1}$
- ${}^nC_r = \frac{n}{r} {}^{n-1}C_{r-1}$

If $a, b \in \mathbb{R}$ and $n \in \mathbb{N}$, then

$$(a+b)^n = {}^nC_0 a^n b^0 + {}^nC_1 a^{n-1} b^1 + {}^nC_2 a^{n-2} b^2 + \dots + {}^nC_n a^0 b^n$$

• **Remarks:** If the index of the binomial is n then the expansion contains $n+1$ terms.

• In each term, the sum of indices of a and b is always n .

• Coefficients of the terms in binomial expansion equidistant from both the ends are equal.

$$(a-b)^n = {}^nC_0 a^n b^0 - {}^nC_1 a^{n-1} b^1 + {}^nC_2 a^{n-2} b^2 - \dots + (-1)^n {}^nC_n a^0 b^n$$

A sequence is said to be a geometric progression or G.P., if the ratio of any term to its preceding term is same throughout. This constant factor is called 'Common ratio'. Usually we denote the first term of a G.P. by 'a' and its common ratio by 'r'.
The general or n^{th} term of G.P. is given by $a_n = ar^{n-1}$
The sum S_n of the first n terms of G.P. is given by
$$S_n = \frac{a(r^n - 1)}{r - 1} \text{ or } \frac{a(1 - r^n)}{1 - r} \text{ if } r \neq 1$$

Sum of infinite terms of G.P. is given by
$$S_n = \frac{a}{1 - r}; |r| < 1$$

An A.P. is a sequence in which terms increase or decrease regularly by the fixed number (same constant). This fixed number is called 'common difference' of the A.P. If 'a' is the first term & 'd' is the common difference and 'l' is the last term of A.P., then general term or the n^{th} term of the A.P. is given by $a_n = a + (n-1)d$.

The sum S_n of the first n terms of an A.P. is given by

$$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}[a + l]$$

If a, b, c are in G.P., b is the GM between a & c, $b^2 = ac$, therefore $b = \sqrt{ac}$; $a > 0, c > 0$
If a, b are two given numbers & $a, G_1, G_2, \dots, G_n, b$ are in G.P., then $G_1, G_2, G_3, \dots, G_n$ are n GMs between a & b.
$$G_1 = a \cdot \left(\frac{b}{a}\right)^{\frac{1}{n+1}}, G_2 = a \cdot \left(\frac{b}{a}\right)^{\frac{2}{n+1}}, \dots, G_n = a \cdot \left(\frac{b}{a}\right)^{\frac{n}{n+1}}$$

or
$$G_1 = ar, G_2 = ar^2, \dots, G_n = ar^{n-1}$$

where $r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$

Sequence is a function whose domain is the set of N natural numbers.

Real sequence: A sequence whose range is a subset of R is called a real sequence

Series: If $a_1, a_2, a_3, \dots, a_n$ is a sequence, then the expression $a_1 + a_2 + a_3 + \dots + a_n + \dots$ is a series. A series is finite or infinite according to the no. of terms in the corresponding sequence as finite or infinite.

Progression: Those sequences whose terms follow certain patterns are called progression.

Sequences and Series

Geometric Mean (GM)

Geometric Progression (G.P.)

Definition

Arithmetic Progression (A.P.)

In a series, where each term is formed by multiplying the corresponding term of an AP & GP is called Arithmetico-Geometric Series.

Eg: $1 + 3x + 5x^2 + 7x^3 + \dots$

Here, $1, 3, 5, \dots$ are in AP and $1, x, x^2, \dots$ are in GP

- If A and G are respectively arithmetic and geometric means between two positive numbers a and b; then the quadratic equation have a, b as its root is $x^2 - 2Ax + G^2 = 0$
- If A and G be the AM & GM between two positive numbers, then the numbers are $A \pm \sqrt{A^2 - G^2}$

Arithmetico Geometric Series

Relationship between AM & GM

Sum upto terms of some special Sequences

Arithmetic Mean (AM)

Properties of AM & GM between two quantities

Let A and G be the AM and GM of two given positive real numbers a & b, respectively. Then,
$$A = \frac{a+b}{2} \text{ and } G = \sqrt{ab}$$

Thus, we have, $A - G = \frac{a+b}{2} - \sqrt{ab} = \frac{a+b-2\sqrt{ab}}{2}$
$$= \frac{(\sqrt{a}-\sqrt{b})^2}{2} \geq 0 \dots\dots\dots (1)$$

From (1), we obtain the relationship $A \geq G$

- Sum of first 'n' natural numbers
$$\sum_{k=1}^n k = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$
- Sum of squares of first n natural numbers
$$\sum_{k=1}^n k^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$
- Sum of cubes of first n natural numbers
$$\sum_{k=1}^n k^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2 = \left(\sum_{k=1}^n k\right)^2$$
- Sum of first 'n' odd natural numbers
$$\sum_{k=1}^n (2k-1) = 1 + 3 + 5 + \dots + (2n-1) = n^2$$

If three numbers are in A.P., then the middle term is called AM between the other two, so if a, b, c, are in A.P., b is AM of a and c.

AM for any 'n' +ve numbers $a_1, a_2, a_3, \dots, a_n$ is

$$AM = \frac{a_1 + a_2 + a_3 + \dots + a_n}{n}$$

n-Arithmetic Mean between Two Numbers: If a, b are any two given numbers & $a, A_1, A_2, \dots, A_n, b$ are in A.P. then $A_1, A_2, \dots, A_n$ are n AM's between a & b.

$$A_1 = a + \frac{b-a}{n+1}, A_2 = a + \frac{2(b-a)}{n+1}, \dots, A_n = a + \frac{n(b-a)}{n+1}$$

or $A_1 = a + d, A_2 = a + 2d, \dots, A_n = a + nd$,
where $d = \frac{b-a}{n+1}$

The distance between the points $A(x_1, y_1)$ and $B(x_2, y_2)$ is

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

1. When two lines of the slope m_1 & m_2 are at right angles, the Product of their slope is -1 , i.e., $m_1 m_2 = -1$. Thus, any line perpendicular to $y = mx + c$ is of the form $y = -\frac{1}{m}x + d$ where d is any parameter.

2. Two lines $ax + by + c = 0$ and $a'x + b'y + c' = 0$ are perpendicular if $aa' + bb' = 0$. Thus, any line perpendicular to $ax + by + c = 0$ is of the form $bx - ay + k = 0$, where k is any parameter.

1. The image of a point (x_1, y_1) about a line $ax + by + c = 0$ is:

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = -\frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}}$$

2. Similarly, foot of perpendicular from a point on the line is:

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{-(ax_1 + by_1 + c)}{\sqrt{a^2 + b^2}}$$

Reflection & foot of perpendicular a point about a line

The length of the perpendicular from $P(x_1, y_1)$ on $ax + by + c = 0$ is:

$$\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Length of the perpendicular from a point on a line

Equation of Straight line in various forms

Section Formula

Straight lines

Perpendicular Lines

Slope Formula

Area of Triangle

Area of triangle whose vertices are (x_1, y_1) , (x_2, y_2) and (x_3, y_3) is $\frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$

1. When two lines are parallel their slopes are equal. Thus, any line parallel to $y = mx + c$ is of the type $y = mx + d$, where d is any parameter
2. Two lines $ax + by + c = 0$ and $a'x + b'x + c' = 0$ are parallel if $\frac{a}{a'} = \frac{b}{b'} \neq \frac{c}{c'}$
3. The distance between two parallel lines with equations $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is $\frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$

Parallel lines

The $P(x, y)$ divided the line joining $A(x_1, y_1)$ and $B(x_2, y_2)$ in the ratio $m:n$, then $x = \frac{mx_2 + nx_1}{m + n}$; $y = \frac{my_2 + ny_1}{m + n}$

Note: • If m/n is +ve, the division is internal, but if m/n is -ve, the division is external.

• If $m = n$, then P is the mid-point of the line segment joining A & B .

1. **POINT-SLOPE FORM:** $y - y_1 = m(x - x_1)$ is the equation of a straight line whose slope is ' m ' and passes through the point (x_1, y_1) .

2. **SLOPE INTERCEPT FORM:** $y = mx + c$ is the equation of a straight line whose slope is ' m ' and makes an intercept c on the y -axis.

3. **TWO POINT FORM:** $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$ is the equation of a straight line which passes through (x_1, y_1) & (x_2, y_2) .

4. **INTERCEPT FORM:** $\frac{x}{a} + \frac{y}{b} = 1$ is the equation of a straight line which makes intercepts a & b on x and y axis respectively.

5. **NORMAL / PERPENDICULAR FORM:** $x \cos \alpha + y \sin \alpha = p$ (where $p > 0, 0 \leq \alpha < 2\pi$) is the equation of a straight line where the length of the perpendicular from origin O on the line is p and this perpendicular makes an angle α with +ve x -axis.

6. **GENERAL FORM:** $ax + by + c = 0$ is the equation of a straight line in general form. In this case, slope of line $= -\frac{a}{b}$

- An ellipse is the set of all points in a plane, the sum of whose distances from two fixed points in the plane is constant.
- The two fixed points are called the 'foci' of the ellipse.
- The midpoint of line segment joining foci is called the 'centre' of the ellipse.
- The line segment through the foci of the ellipse is called 'major axis'.
- The line segment through centre & perpendicular to major axis is called minor axis.
- The end point of the major axis are called the vertices of the ellipse.
- The equation of ellipse with 'foci' on the x-axis is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
- Length of the latus rectum of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{2b^2}{a}$
- The eccentricity of an ellipse is the ratio of distances from centre of ellipse to one of foci and to one of the vertices of ellipse i.e., $e = \frac{c}{a}$

A circle is a set of all points in a plane that are equidistant from a fixed point in the plane. The fixed point is called the 'centre' of the circle and the distance from the centre to a point on the circle is called the 'radius' of the circle.

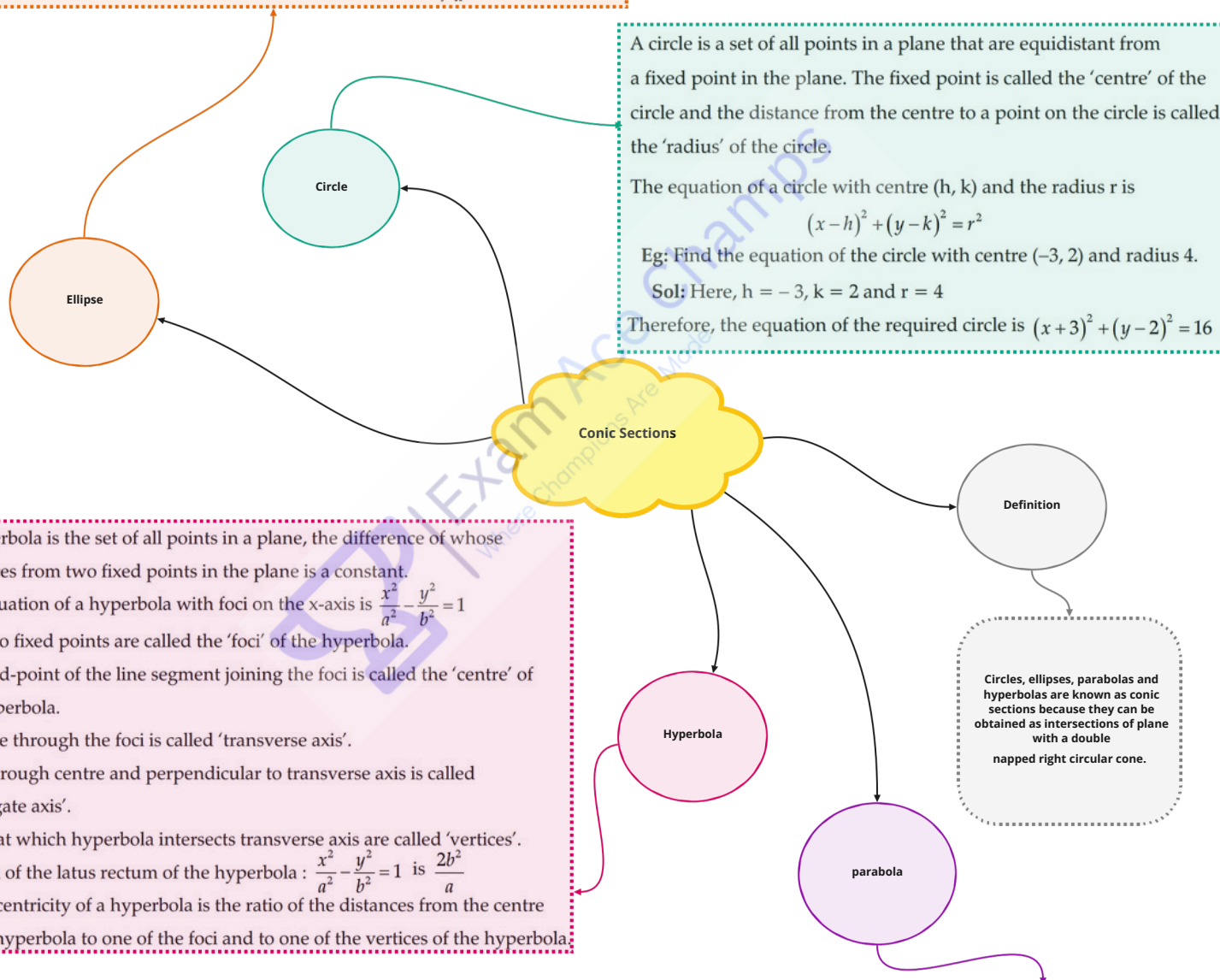
The equation of a circle with centre (h, k) and the radius r is

$$(x-h)^2 + (y-k)^2 = r^2$$

Eg: Find the equation of the circle with centre (-3, 2) and radius 4.

Sol: Here, h = -3, k = 2 and r = 4

Therefore, the equation of the required circle is $(x+3)^2 + (y-2)^2 = 16$



- A hyperbola is the set of all points in a plane, the difference of whose distances from two fixed points in the plane is a constant.
- The equation of a hyperbola with foci on the x-axis is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
- The two fixed points are called the 'foci' of the hyperbola.
- The mid-point of the line segment joining the foci is called the 'centre' of the hyperbola.
- The line through the foci is called 'transverse axis'.
- Line through centre and perpendicular to transverse axis is called 'conjugate axis'.
- Points at which hyperbola intersects transverse axis are called 'vertices'.
- Length of the latus rectum of the hyperbola : $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{2b^2}{a}$
- The eccentricity of a hyperbola is the ratio of the distances from the centre of the hyperbola to one of the foci and to one of the vertices of the hyperbola.

- A parabola is the set of all points in a plane that are equidistant from a fixed line and a fixed point in the plane. Fixed line is called 'directrix' of parabola. Fixed point F is called the 'focus'. A line through focus & perpendicular to directrix is called 'axis'. Point of intersection of parabola with axis is called 'vertex'.

- The equation of parabola with focus at (a, 0), a > 0 and directrix x = -a is $y^2 = 4ax$, where 4a is the length of the latus rectum

Eg: Find the equation of the parabola with vertex at (0, 0) and focus at (0, 2).

Sol: Since, vertex is at (0, 0) and focus is at (0, 2) which lies on y-axis, the y-axis is the axis of parabola. Therefore, equation of the parabola is of the form $x^2 = 4ay$. Thus we have $x^2 = 4(2)y$
i.e., $x^2 = 8y$

The coordinates of the centroid of the triangle, whose vertices are (x_1, y_1, z_1) , (x_2, y_2, z_2) and (x_3, y_3, z_3) are $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3}\right)$

Eg: The centroid of a triangle ABC is at the point (1, 1, 1). If the coordinates of A and B are (3, -5, 7) and (-1, 7, -6), respectively, find the coordinates of the point C.

Sol: Let the coordinates of C be (x, y, z) and the coordinates of the centroid G be (1, 1, 1). Then $\frac{x+3-1}{3} = 1$, i.e., $x=1$;

$$\frac{y-5+7}{3} = 1, \text{ i.e., } y=1;$$

$$\frac{z+7-6}{3} = 1, \text{ i.e., } z=2. \text{ So, } C(x, y, z) = (1, 1, 2)$$

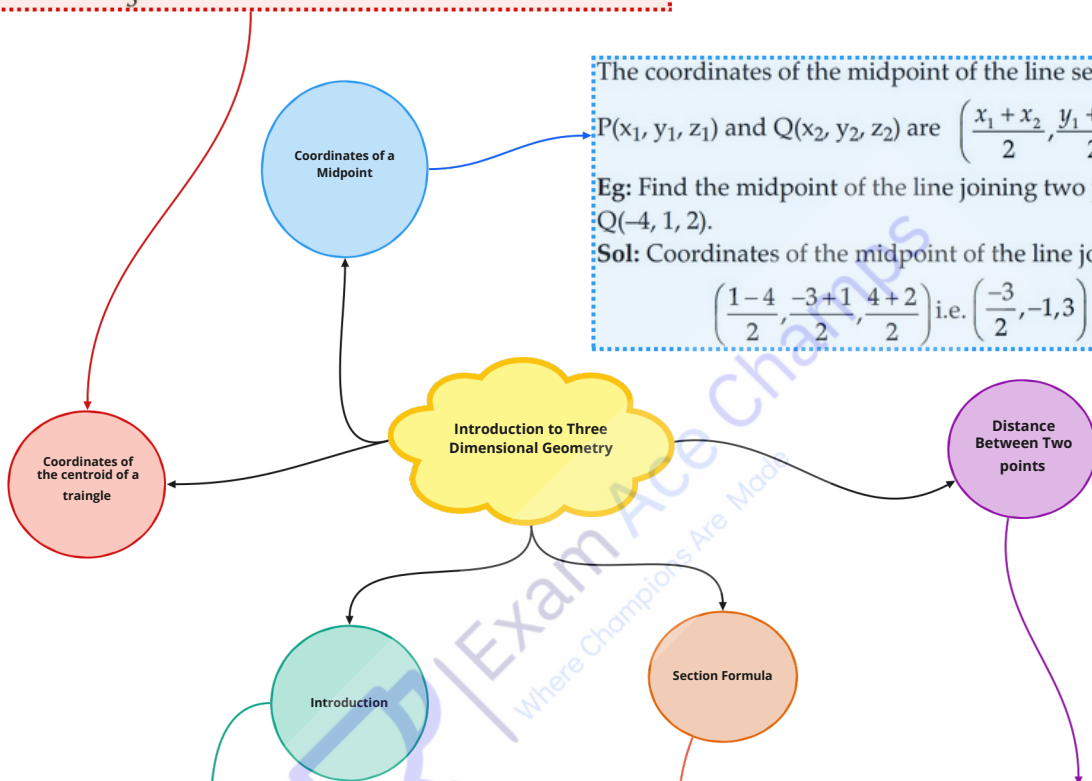
The coordinates of the midpoint of the line segment joining two points

$$P(x_1, y_1, z_1) \text{ and } Q(x_2, y_2, z_2) \text{ are } \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right).$$

Eg: Find the midpoint of the line joining two points P(1, -3, 4) and Q(-4, 1, 2).

Sol: Coordinates of the midpoint of the line joining the points P & Q are

$$\left(\frac{1-4}{2}, \frac{-3+1}{2}, \frac{4+2}{2}\right) \text{ i.e. } \left(\frac{-3}{2}, -1, 3\right)$$



- In three dimensions, the coordinate axes of a rectangular cartesian coordinate system are three mutually perpendicular lines. The axes are called x, y and z axes.
- The three planes determined by the pair of axes are the coordinate planes, called xy, yz and zx-planes.
- The three coordinate planes divide the space into eight parts known as octants.
- The coordinates of a point P in 3D Geometry is always written in the form of triplet like (x, y, z) . Here, x, y and z are the distances from yz, zx and xy planes, respectively.

Eg:

- Any point on x-axis is : $(x, 0, 0)$
- Any point on y-axis is : $(0, y, 0)$
- Any point on z-axis is : $(0, 0, z)$

Distance between two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is given by

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Eg: Find the distance between the points P(1, -3, 4) and (-4, 1, 2).

Sol: The distance PQ between the points P & Q is given by

$$PQ = \sqrt{(-4-1)^2 + (1+3)^2 + (2-4)^2} = \sqrt{25+16+4} = \sqrt{45} = 3\sqrt{5} \text{ units}$$

The coordinates of the point R which divides the line segment joining two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ internally and externally in the ratio $m : n$ are given by

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}, \frac{mz_2 + nz_1}{m+n}\right) \quad \& \quad \left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n}, \frac{mz_2 - nz_1}{m-n}\right)$$

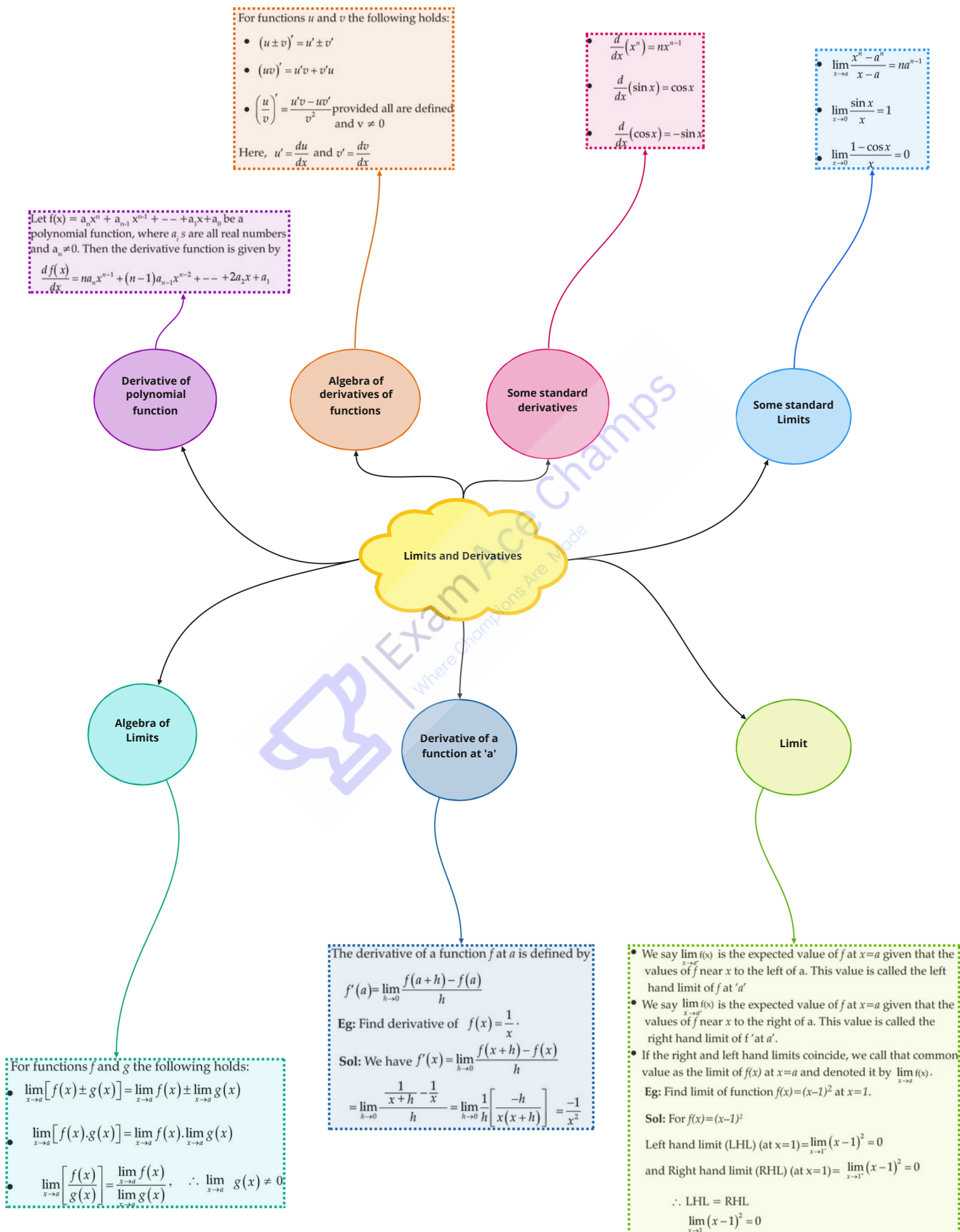
respectively.

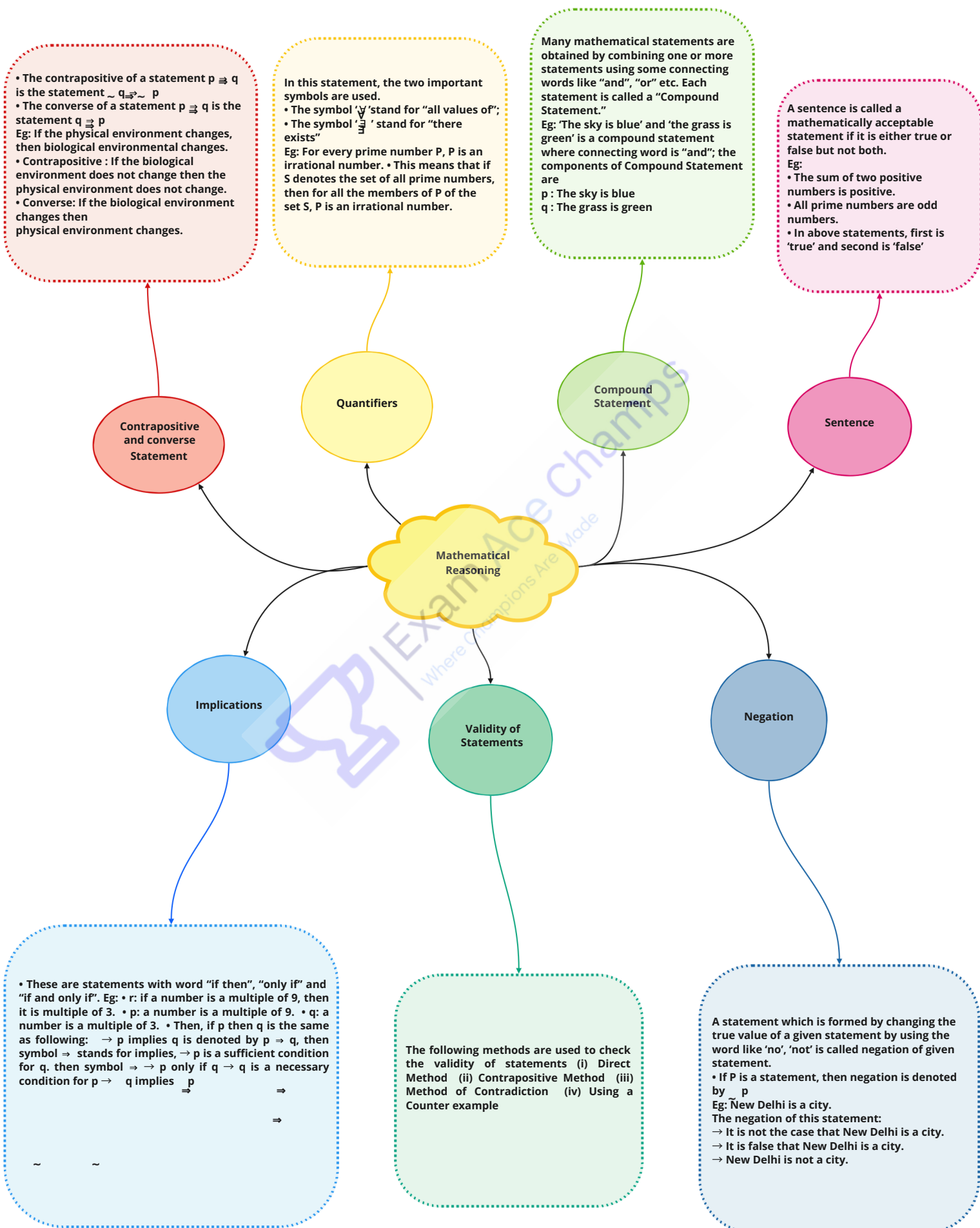
Eg: Find the coordinates of the point which divides the line segment joining the points (1, -2, 3) and (3, 4, -5) in the ratio 2:3 internally.

Sol : Let $P(x, y, z)$ be the point which divides line segment joining A (1, -2, 3) and B (3, 4, -5) internally in the ratio 2:3. Therefore,

$$x = \frac{2(3) + 3(1)}{2+3} = \frac{9}{5} \quad y = \frac{2(4) + 3(-2)}{2+3} = \frac{2}{5} \quad z = \frac{2(-5) + 3(3)}{2+3} = \frac{-1}{5}$$

Thus, the required point is $\left(\frac{9}{5}, \frac{2}{5}, \frac{-1}{5}\right)$.





$$\bullet \text{ M.D.}(\bar{x}) = \frac{1}{N} \sum f_i |x_i - \bar{x}|$$

$$\text{where } \bar{x} = a + \frac{\sum_{i=1}^n f_i d_i}{N} \times h$$

$$\text{and } d_i = \frac{x_i - a}{h}$$

$$\text{Here, } a = \text{assumed mean}$$

$$h = \text{common factor}$$

$$N = \text{sum of frequencies}$$

$$\bullet \text{ M.D.}(M) = \frac{1}{N} \sum f_i |x_i - M|$$

$$\text{where, } M(\text{median}) = l + \frac{\frac{N}{2} - C}{f} \times h$$

Variance

$$(\sigma^2) = \frac{h^2}{N^2} \left[N \sum f_i y_i^2 - (\sum f_i y_i)^2 \right]$$
Standard Deviation

$$\sigma = \frac{h}{N} \sqrt{N \sum f_i y_i^2 - (\sum f_i y_i)^2}$$

$$\text{where } y_i = \frac{x_i - A}{h}$$

$$N = \sum f_i$$

$$A = \text{assumed mean}$$

$$h = \text{width of class interval}$$

Variance of a discrete frequency distribution

$$\text{Var}(\sigma^2) = \frac{1}{N} \sum f_i (x_i - \bar{x})^2$$
Standard Deviation (S.D.) of a discrete frequency distribution

$$\text{S.D.}(\sigma) = \sqrt{\frac{1}{N} \sum f_i (x_i - \bar{x})^2}$$

$$\text{where } N = \sum f_i \text{ and } \bar{x} = \text{mean}$$

Variance of ungrouped data

$$\text{Var}(\sigma^2) = \frac{1}{n} \sum (x_i - \bar{x})^2$$
Standard Deviation (S.D.) for ungrouped data:

$$\text{S.D.}(\sigma) = \sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2}$$

Variance of a continuous frequency distribution

$$\text{Var}(\sigma^2) = \frac{1}{N} \sum f_i (x_i - \bar{x})^2$$
Standard deviation of a continuous frequency distribution

$$\text{S.D.}(\sigma) = \frac{1}{N} \sqrt{N \sum f_i x_i^2 - (\sum f_i x_i)^2}$$

Step deviation or shortcut Method

Shortcut method to find variance and standard deviation

Variance & standard Deviation of a discrete frequency distribution

Variance and standard Deviation for ungrouped data

Variance & standard deviation of a continuous frequency distribution

Median Class

Relationship between AM & GM

Range

Coefficient of variation

Mean Deviation

Coefficient of variation (C.V.)

$$= \frac{\sigma}{\bar{x}} \times 100; \bar{x} \neq 0$$

$$\text{where } \bar{x} \text{ is mean}$$

Median class is the class interval whose C.F. is just greater than or equal to $\frac{N}{2}$, f , h and C are respectively the lower limit, the frequency, width of the median class and C the cumulative frequency (C.F.) of class just preceding to median class.

The degree to which numerical data tend to spread about an average value is called the dispersion of the data. The four dispersions are: Range, Mean deviation, Standard deviation and variance.

The difference between the highest and the lowest element of a data called its range.

$$\text{range} = x_{\max} - x_{\min}$$

Eg: Find the range of the given data:
4, 7, 8, 9, 10, 12, 13, 17
Here, $x_{\max} = 17$ and $x_{\min} = 4$
Range = $17 - 4 = 13$

Mean Deviation (M.D.) for ungrouped data

$$\bullet \text{ M.D. about mean i.e., } \text{M.D.}(\bar{x}) = \frac{\sum |x_i - \bar{x}|}{n}, \text{ where } \bar{x} \text{ is mean and } n = \text{no. of terms}$$

$$\bullet \text{ M.D. about median i.e., } \text{M.D.}(M) = \frac{\sum |x_i - M|}{n}, \text{ where } M \text{ is median and } n = \text{no. of terms}$$

Mean Deviation (M.D.) for grouped data

$$\text{M.D. about mean i.e., } \text{M.D.}(\bar{x}) = \frac{\sum f_i |x_i - \bar{x}|}{N}$$

$$\text{M.D. about median i.e., } \text{M.D.}(M) = \frac{1}{N} \sum f_i |x_i - M|$$

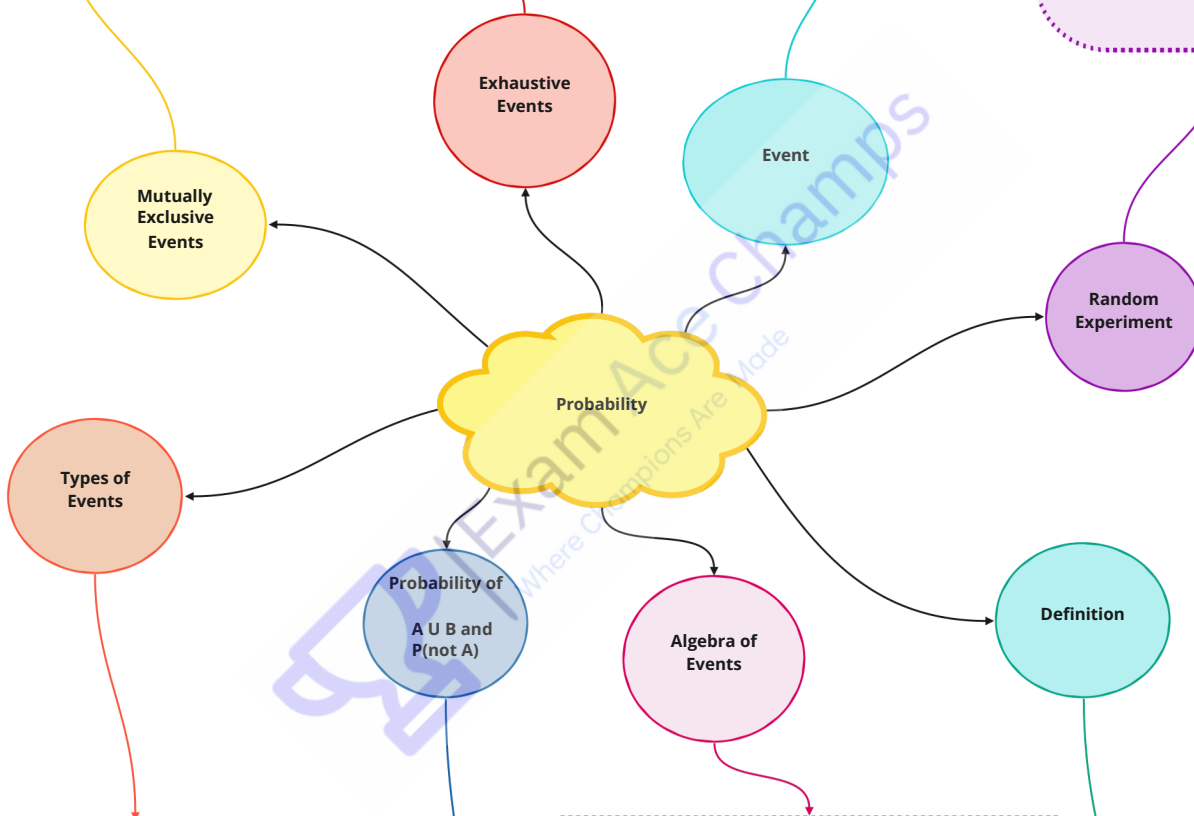
$$\text{where } N = \sum f_i$$

Events A & B are called mutually exclusive events if occurrence of any one of them excludes occurrence of other event, i.e. they cannot occur simultaneously.
Eg: A die is thrown. Event A = All even outcomes & event B = All odd outcomes. Then A & B are mutually exclusive events, they cannot occur simultaneously.
Note: Simple events of a sample space are always mutually exclusive.

Many events that together form sample space are called exhaustive events.
Eg: A die is thrown. Event A = All even outcomes and event B = All odd outcomes. Event A & B together forms exhaustive events as it forms sample space

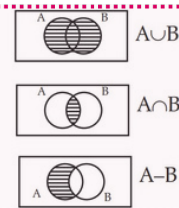
It is the set of favourable outcomes. Any subset E of a sample space S is called an event.
Eg: Event of getting an even number (outcome) in a throw of a die.
Occurrence of event: The event E of a sample space 'S' is said to have occurred if the outcome w of the experiment is such that $w \in E$. If the outcome w is such that $w \notin E$ we say that event E has not occurred.

An Experiment is called random experiment if it satisfies the following two conditions:
• It has more than one possible outcome.
• It is not possible to predict the outcome in advance.
Outcome: A possible result of a random experiment is called its outcome.
Sample Space: Set of all possible outcomes of a random experiment is called sample space. It is denoted by symbol 'S'. Eg: In a toss of a coin, sample space is Head & Tail i.e., $S = \{H, T\}$
Sample Point: Each element of the Sample Space is called a sample point.
Eg: In a toss of a coin, head is a sample point
Equally Likely Outcomes: All outcome with equal probability.

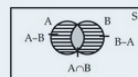


• Impossible and Sure Event: The empty set ϕ is called an Impossible event, whereas the whole sample space 'S' is called 'Sure event'.
Eg: In a rolling of a die, impossible event is that number more than 6 and event of getting number less than or equal to 6 is sure event.
• Simple Event: If an event has only one sample point of a sample space, it is called a 'simple event'.
Eg: In rolling of a die, simple event could be the event of getting number 4.
• Compound Event: If an event has more than one sample point, it is called a 'compound event'.
Eg: In rolling of a die, compound event could be event of getting an even number.
• Complementary Event: Complement event to A = 'not A'
Eg: If an event A = Event of getting odd number in a throw of a die i.e., $\{1, 3, 5\}$ then, complementary event to A = Event of getting an even number in a throw of a die, i.e. $\{2, 4, 6\}$ $A' = \{W: W \in S \text{ and } W \notin A\} = S - A$ (where S is the sample space) $\in \notin$

• Event A or B or $(A \cup B)$
 $A \cup B = \{w: w \in A \text{ or } w \in B\}$
• Event A and B or $(A \cap B)$
 $A \cap B = \{w: w \in A \text{ and } w \in B\}$
• Event A but not B or $(A - B)$
 $A - B = A \cap B'$



If A and B are any two events, then
• $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
• $P(A \cap B) = P(A) + P(B) - P(A \cup B)$
If A and B are mutually exclusive, then $P(A \cup B) = P(A) + P(B)$
• Probability of the event 'not A'
 $P(A') = P(\text{not } A) = 1 - P(A)$



Probability is the measure of uncertainty of various phenomenon, numerically. It can have positive value from 0 to 1.

$$\text{Probability} = \frac{\text{No. of favourable outcomes}}{\text{Total no. of outcomes}}$$

Eg: Probability of getting an even no. in a throw of a die.

Sol. Here, favourable outcomes = $\{2, 4, 6\}$
Total no. of outcomes = $\{1, 2, 3, 4, 5, 6\}$

$$\text{Probability} = \frac{3}{6} = \frac{1}{2}$$



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